Start: 2:50 pm

- If research question is clear or reasonable
    - We have a "Gender" column - patients reported → chance that they have used the terms "sex" and "gender" interchangeably so let's use "sex"!
        - Sex is more biologically relevant
    - Overall: good research topic as discussed before - just make sure when you put it in the assignment is well-written

1. Does gender modify the relationship between gastric disease histopathological stage and gastric microbiome composition and function?

- Be clear on what you want to investigate and whether it's related to the microbiota

- Sequencing depth range is still within reasonable range (after denoising) (253)
- 253 bp truncation length is common for 16S rRNA data
- Choices from curves, including rarefaction, are good choices
    - 3145 rarefaction sampling depth value is good

- Additional filtering:
    - Up to us if we want to filter out ASVs below 0.5% appearance
        - Should write out why we left them in or took them out
    - Do not filter based on metadata at this step (can do it in R later)
    - Do not need any diversity metrics from QIIME

Next steps:

- Export files into R
- Preliminary analysis (alpha and beta diversity)
- Start thinking about questions in more detail → what will the aims be and what types of statistical tests/metrics we want to use
- PERMANOVA for H.pylori (refer to Meeting 2)
- Look at indicator species for each stage/differential abundance/functional

Aims:

"Question is still broad so she recommends just doing random shit and hopefully finding something interesting"

1. Module 13 - alpha/beta diversity
    a. Comparing the different stages of gastric cancer progress with respect to sex (?)
    b. Diversity metrics broken down by stages with different between sexes

    c. Diversity at different stages then stratify it by sex
2. Looking at the taxonomic differences between the stages/groups → indicator species
3. Going into functional analysis (if you see differences, maybe there are:)
    a. Pathways that relates to inflammation
    b. Sex differences where one sex is doing better
    c. Metabolism, enzyme, nitty gritty biologic function (k***egg*** terms)
4. No random forest